WORKBOOK · SLIDES 1 – 75

Applications & Solutions

Every application and worked example from the course. Each application is on its own page, and the solution is on the next page: a course reminder, the step-by-step solution in the professor's method, and the rules to remember.

28	4	28
APPLICATIONS	PARTS	RULE BOXES

- Solution taken from the slides (2)
- Slide solution + full steps added (4)
- No solution in the slides: solved with the course method (22)

How to use it: solve the application on its page, then turn the page to compare. Read the ★ rules box last. These are the reflexes the exam expects.

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PART 1 · SLIDES 3 – 19

Introduction to Descriptive Statistics

Vocabulary: population, sample, unit, variable, modalities, counts and frequencies.

1 Frequency and count: phone colours

Slide 10

APPLICATION

6 students have black phones → $n_i = \dots$
If 10 students are surveyed → $f_i = \dots$ or $\dots$ %

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 1

1 Frequency and count: phone colours

Slide 10 | Solved with the course method

COURSE REMINDER

When we observe a variable on N units, we group the units by **modality** (category or value).

- **Count** (absolute frequency) n_i: the number of units that have modality i.
- **Relative frequency** $f_i = \dfrac{n_i}{N}$ with $N = \sum n_i$: the **share** of units with modality i. Multiply it by 100 to get a percentage.

SOLUTION

The **count** is the number of units having the modality "black": $n_i = 6$.

The **relative frequency** divides the count by the total:

$$f_i = \frac{n_i}{N} = \frac{6}{10} = 0.6 \quad \text{or} \quad 60\,\%$$

RULES TO REMEMBER

- Count n_i = **how many** units take modality i (a whole number).
- Relative frequency $f_i = \dfrac{n_i}{N}$ with $N = \sum n_i$; it is between 0 and 1, and $f_i \times 100$ gives the %.
- Check: $\sum n_i = N$ and $\sum f_i = 1$ (100 %).

2 Multiple choice (QCM)

Slide 11

APPLICATION

Q1. What is the main goal of descriptive statistics?
A. Predict future data outcomes · B. Describe a large amount of data concisely and meaningfully · C. Identify cause-effect relationships · D. Infer conclusions from samples to populations

Q2. Which of the following is not a feature of descriptive statistics?
A. Summarizing data using graphs · B. Making predictions based on probabilities · C. Calculating frequencies and averages · D. Organizing data in tables

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 2

2 Multiple choice (QCM)

Slide 11 | Solved with the course method

COURSE REMINDER

Descriptive statistics is the branch of statistics that **summarizes and describes** the essential features of a dataset, using tables, numerical indicators and graphs.

Statistics has **two branches**:

- **Descriptive statistics**: collection, organization, summary and presentation of data.
- **Inferential statistics**: uses **probability theory** to make predictions or decisions about a population from a sample.

SOLUTION

Q1 → B. Descriptive statistics *summarizes and describes* the essential features of a dataset using tables, indicators and graphs. A and D are **inferential** statistics. C (cause and effect) is not covered by descriptive statistics.

Q2 → B. Making predictions with probability theory is **inferential** statistics. A, C and D are what descriptive statistics does: collect, organize, summarize and present data.

RULES TO REMEMBER

- **Descriptive** = collect, organize, summarize, present (tables, numbers, graphs).
- **Inferential** = use probability to predict or decide about a population from a sample.
- If an answer contains the words *predict, infer, generalize*, it is inferential, not descriptive.

3 True or False: core concepts

Slide 12

APPLICATION

Read each statement and mark it as TRUE or FALSE.

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 3

3 True or False: core concepts

Slide 12 | Solved with the course method

COURSE REMINDER

- **Population**: all the units concerned by the study. **Sample**: a subset of the population. **Statistical unit**: one element that is observed.
- **Variable (character)**: a feature measured on each unit. It is **qualitative** (nominal = no order, ordinal = ordered) or **quantitative** (discrete = countable, continuous = measurable).
- **Modalities**: the values a variable can take. They obey two rules: **incompatibility** (a unit cannot belong to two modalities) and **exhaustiveness** (every unit fits one modality).

SOLUTION

#	Statement	Answer	Why
1	A “population” is a subset of the sample.	FALSE	It is the reverse: the **sample** is a subset of the **population**.
2	A “unit” is one element from the population.	TRUE	The statistical unit is the element that is observed (e.g. a student).
3	Variables can be both qualitative and quantitative.	TRUE*	Both kinds of variables exist, and a study can contain both. **But one given variable is either qualitative or quantitative, never both at the same time.*
4	An “ordinal variable” has non-ranked modalities.	FALSE	Ordinal means **ordered** (e.g. education level). Unordered modalities are **nominal**.
5	A “discrete variable” can take an infinite number of values.	FALSE	Discrete means finite or **countable** values (e.g. number of children). Measurable values in an interval are **continuous**.
6	A census is always preferable to a survey because it provides more accurate data.	FALSE	A census is more accurate, but it is **costly and slow**. A representative sample is often the better choice.
7	A character must be numeric to be studied statistically.	FALSE	Qualitative characters (eye colour, nationality...) are also studied, using counts and frequencies.
8	A modality of a variable must be both exhaustive and mutually exclusive.	TRUE	These are the two rules: **incompatibility** and **exhaustiveness**.
9	A statistical unit can have more than one modality for the same variable.	FALSE	Incompatibility rule: a unit belongs to **one** modality only.
10	A variable can be both qualitative and continuous.	FALSE	Continuous is a sub-type of **quantitative**. A qualitative variable can only be nominal or ordinal.

★ RULES TO REMEMBER

- Population ⊃ sample ⊃ unit (whole group → selected part → one element).
- Qualitative → **nominal** (no order) or **ordinal** (order). Quantitative → **discrete** (countable) or **continuous** (measurable).
- Modalities must be **incompatible** (one modality per unit) and **exhaustive** (every unit fits somewhere).
- Statements with *always* or *must* are often FALSE. Check them carefully.

4 Laptop survey: identify the key elements

Slide 13

APPLICATION

A university is analyzing the types of laptops used by its **1,200 students**. A survey is conducted on **150 randomly selected students**, recording their laptop brands (HP, Dell, Apple, Asus, Others). Identify the population, sample, statistical unit, variable, modalities and nature of the variable.

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 4

4 Laptop survey: identify the key elements

Slide 13 | Solved with the course method

COURSE REMINDER

Every statistical study is described by the same key elements:

- **Population**: the whole set of units under study. **Sample**: the subset that is actually observed.
- **Statistical unit**: the element observed. **Variable**: the characteristic measured on it. **Modalities**: its possible values.
- **Nature** of the variable: qualitative (nominal/ordinal) or quantitative (discrete/continuous).
- **Simple random sampling**: each unit of the population has the same chance of being selected.

SOLUTION

Element	Answer
Population	The 1,200 students of the university
Sample	The 150 randomly selected students (simple random sampling)
Statistical unit	One student
Variable	The laptop brand
Modalities	HP, Dell, Apple, Asus, Others
Nature	**Qualitative nominal** (the brands are not numbers and have no natural order)

RULES TO REMEMBER

- Population = **everybody** concerned; sample = the people **actually asked**.
- Unit = **who or what** you observe; variable = **what you measure** on it; modalities = the **possible answers**.
- To find the nature: is it a number? No → qualitative. Can you rank the answers? No → nominal.
- "Others" is there to make the modalities **exhaustive**.

5 Complete the table: variable types

Slide 14

APPLICATION

Give the type and subtype of: eye colour, monthly salary, level of satisfaction (low/medium/high), number of children in a family, blood type (A, B, AB, O), height in cm.

YOUR WORK

Solution on the next page →

5

SOLUTION · APPLICATION 5

Complete the table: variable types

Slide 14 | Solved with the course method

COURSE REMINDER

Type	Subtype	Meaning	Example
Qualitative (non-numerical)	Nominal	categories with no order	eye colour, gender
	Ordinal	ordered categories	level of education
Quantitative (numerical)	Discrete	finite / countable values	number of children
	Continuous	measurable values in an interval	height, weight, income

SOLUTION

Variable	Type	Subtype	Reason
Eye color	Qualitative	Nominal	categories with no order
Monthly salary	Quantitative	Continuous	measurable amount (any value in an interval)
Level of satisfaction	Qualitative	Ordinal	low < medium < high: ordered
Number of children	Quantitative	Discrete	countable: 0, 1, 2, ...
Blood type	Qualitative	Nominal	A, B, AB, O have no ranking
Height in cm	Quantitative	Continuous	measurable

RULES TO REMEMBER

- Ask “can I **count** it?” → discrete. Ask “can I **measure** it?” → continuous.
- Letters or words with a natural ranking → ordinal. Without a ranking → nominal.
- Money, height, weight, time and age are treated as **continuous** in this course.

6 Find the errors in a table

Slide 15

APPLICATION

Age of student	Number of students
[18,20[	5
[20,21[& [21,22[	3
[22,24[	7
[24,28[	4
> 200	1

YOUR WORK

Solution on the next page →

6

SOLUTION · APPLICATION 6

Find the errors in a table

Slide 15 | Solved with the course method

COURSE REMINDER

A correct **frequency table** respects these rules:

- For a continuous variable, the data are grouped into classes $[e_{i-1}, e_i[$ that follow each other with no gap and no overlap.
- **Incompatibility**: each observation belongs to one class only. **Exhaustiveness**: every observation fits a class.
- Class width $a_i = e_i - e_{i-1}$. When the widths are unequal, use the **corrected frequency** $n_i^c = \frac{n_i}{a_i} \times a_0$, where a_0 is the smallest width.

SOLUTION

1. **[20,21[& [21,22[in one row.** Two classes are mixed into one line. Write one class **[20,22[**, or give two rows with their own counts.
2. **"> 200".** An age above 200 years is impossible, so it is a data entry error. The class also leaves a gap between 28 and 200, so the classes are no longer **contiguous** or **exhaustive**. Use a realistic last class such as [28, 32[or "28 and more".
3. **Unequal widths** (2, 2, 2, 4). This is allowed, but [24,28[is twice as wide, so its height in a histogram must use the **corrected frequency**: $n_i^c = n_i / k_i$ with $k_i = a_i / a_0 = 4/2 = 2$.
4. **No Total row.** A frequency table should end with $N = \sum n_i$ (here 20).

RULES TO REMEMBER

- Classes are written $[e_{i-1}, e_i[$: they follow each other with **no gap and no overlap**.
- One row = one modality or class (incompatibility rule).
- Values must be **realistic**. Every observation must fall in exactly one class (exhaustiveness).
- Unequal widths → corrected frequency $n_i^c = \frac{n_i}{a_i} \times a_0$, where a_0 is the smallest width.

7 Census or sample?

Slide 16

APPLICATION

Scenario A: a university wants to find the most common laptop brand among its 100,000 alumni.
Scenario B: a manager wants to determine the salary structure of the 25 employees in his department.
Where is a census more appropriate? Where is a sample more practical? Explain.

YOUR WORK

Solution on the next page →

7

SOLUTION · APPLICATION 7

Census or sample?

Slide 16 | Solved with the course method

COURSE REMINDER

There are two ways to collect data:

- A **census** (French: *recensement*) studies the **entire population**: every unit is observed. It is **more accurate**, but **costly and time-consuming**.
- **Sampling** (a survey) studies only a **subset** of the population. It is **less expensive and quicker**, but the sample must be **representative**.

Types of sampling: quota sampling, simple random sampling, stratified sampling, judgement (reasoned) sampling...

SOLUTION

	Scenario A (100,000 alumni)	Scenario B (25 employees)
Best choice	**Sample**	**Census**
Why	Asking 100,000 people is costly and slow. A representative sample (e.g. simple random or stratified) gives a reliable answer quickly and cheaply.	The population is very small, so asking all 25 is cheap and fast. A census is also **more accurate**: it has no sampling error.

RULES TO REMEMBER

- **Census**: full population → more accurate, but costly and time-consuming.
- **Sampling**: a subset → cheaper and quicker, but it must be **representative**.
- Large population → sample. Small, reachable population → census.
- Sampling types: quota, simple random, stratified, judgement (reasoned).

8 Match the concepts with their definitions

Slide 17

APPLICATION

Match: Population, Statistical unit, Variable, Modality, Sample with the definitions (specific value taken by a variable / subset selected from a larger group / entire group under study / characteristic measured on each element / individual element of the group).

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 8

8 Match the concepts with their definitions

Slide 17 | Solved with the course method

COURSE REMINDER

The five basic words of statistics, from the biggest to the most specific:

Population → Sample → Statistical unit → Variable → Modality

A population is made of units, and a sample takes some of them. On each unit we measure a variable, and the answer we get is a modality.

SOLUTION

Term	Correct definition
Population	Entire group under study
Statistical unit	Individual element of the group
Variable	A characteristic measured on each element
Modality	A specific value taken by a variable
Sample	Subset selected from a larger group

RULES TO REMEMBER

- Memory chain: **Population → Sample → Unit → Variable → Modality** (from the biggest to the most specific).

9 Coffee consumption study

Slide 18

APPLICATION

A marketing agency studies the coffee consumption habits of employees in tech startups across France. It selects 200 employees from various companies to answer questions like age, job title, cups of coffee per day and coffee brand preference.

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 9

9 Coffee consumption study

Slide 18 | Solved with the course method

COURSE REMINDER

To describe a survey, first separate **who** is concerned (population), **who answered** (sample), and **what was measured** (the variables).

- **Qualitative variable**: the answer is a word or category (job title, brand).
- **Quantitative variable**: the answer is a number. It is **discrete** if you count it (0, 1, 2, ...) and **continuous** if you measure it (age, height, money).

SOLUTION

1. **Population:** all employees of tech startups in France.
2. **Sample:** the 200 employees who were selected and answered.
3. **Qualitative variables:** job title (nominal) and coffee brand preference (nominal).
4. **Quantitative variables:** cups of coffee per day (discrete) and age (continuous).
5. **Cups of coffee per day is discrete**, because it takes countable whole values (0, 1, 2, 3, ...). You count cups, you do not measure them; a value like 2.37 cups is not observed.

RULES TO REMEMBER

- The population is defined by the **scope** of the study (who? where?), not by the people asked.
- "Number of ..." or "... per day" that you count → **discrete**.
- Always **justify** the type with the words *countable* or *measurable*.

10 Languages spoken: compute frequencies

Slide 19

APPLICATION

Languages spoken	Count n_i	Relative frequency f_i
1	6	?
2	10	?
3	4	?
Total	**20**	**1.00**

YOUR WORK

Solution on the next page →

10

SOLUTION · APPLICATION 10

Languages spoken: compute frequencies

Slide 19 | Solved with the course method

COURSE REMINDER

The **relative frequency** of a modality is its count divided by the total number of observations:

$$f_i = \frac{n_i}{N}, \qquad N = \sum n_i, \qquad \sum f_i = 1$$

It answers the question "what **proportion** of the units has this modality?".

SOLUTION

Use $f_i = n_i/N$ with $N = 20$:

Languages spoken	n_i	$f_i = n_i/N$	%
1	6	6/20 = **0.30**	30 %
2	10	10/20 = **0.50**	50 %
3	4	4/20 = **0.20**	20 %
Total	**20**	**1.00**	**100 %**

RULES TO REMEMBER

- $f_i = n_i/N$, and always check that the column sums to 1.
- Interpretation sentence: "50 % of the surveyed students speak 2 languages."

PART 2 · SLIDES 20 – 38

Statistical Series Presentation

Tables (counts, frequencies, cumulative frequencies) and graphs for discrete, continuous and qualitative variables.

11 Number of children per household

Slide 26 – 27

APPLICATION

Let there be 10 households with 1, 2, 3, 4 children (n_i = 3, 4, 2, 1). Complete the table and determine the proportion of households with **more than 2 children** and with **fewer than or equal to 3 children**.

YOUR WORK

Solution on the next page →

11

SOLUTION · APPLICATION 11

Number of children per household

Slide 26 – 27 | Slides + full steps added

COURSE REMINDER

For a quantitative variable, values are ordered, so we can **accumulate** the counts:

- Ascending cumulative count $n_i^c \uparrow$: the number of units with $X \leq x_i$. Descending count $n_i^D \downarrow = N - n_i^c$: units with $X > x_i$.
- Cumulative frequency $F^C(x) = F(x) = P(X \leq x)$, also called the **distribution function**. Descending frequency $F^D(x) = P(X > x) = 1 - F(x)$.
- Properties: $0 \leq F(x) \leq 1$, $P(a \leq X \leq b) = F(b) - F(a)$, $P(X > x) = 1 - F(x)$.

SOLUTION

Children x_i	n_i	f_i	$n_i^c \uparrow$	$n_i^D \downarrow$	$F^C \uparrow$	$F^D \downarrow$
1	3	0.3	3	7	0.3	0.7
2	4	0.4	7	3	0.7	0.3
3	2	0.2	9	1	0.9	0.1
4	1	0.1	10	0	1.0	0.0
Total	**10**	**1.0**				

How each column is built: n_i^c adds the counts from the top (3, 3+4=7, ...). $n_i^D = N - n_i^c$ (10−3=7, ...). F^C adds the f_i, and $F^D = 1 - F^C$.

Answers:

- More than 2 children: $P(X > 2) = F^D(2) = 1 - F(2) = 1 - 0.7 = \mathbf{0.3}$ → 30 % of households.
- At most 3 children: $P(X \leq 3) = F^C(3) = \mathbf{0.9}$ → 90 % of households.

```
R (professor's code): X=c(1,1,1,2,2,2,2,3,3,4); T=table(X); V=c(T)
data.frame(Eff=V, EffCum=cumsum(V), Freq=V/sum(V), FreqCum=cumsum(V/sum(V)))
```

RULES TO REMEMBER

- $F^C(x) = F(x) = P(X \leq x)$ is the ascending cumulative frequency (the **distribution function**).
- $F^D(x) = P(X > x) = 1 - F(x)$. The last line of F^C is always 1 and the last line of F^D is always 0.
- "at most / ≤ / fewer than or equal" → read F^C. "more than / >" → read F^D.
- $P(a \leq X \leq b) = F(b) - F(a)$, with $0 \leq F(x) \leq 1$.

12 Plot the three graphs of the previous application

Slide 28

APPLICATION

Plot the three graphs related to the previous application: count bar chart, frequency bar chart and cumulative graph (integral diagram).

YOUR WORK

Solution on the next page →

12

SOLUTION · APPLICATION 12

Plot the three graphs of the previous application

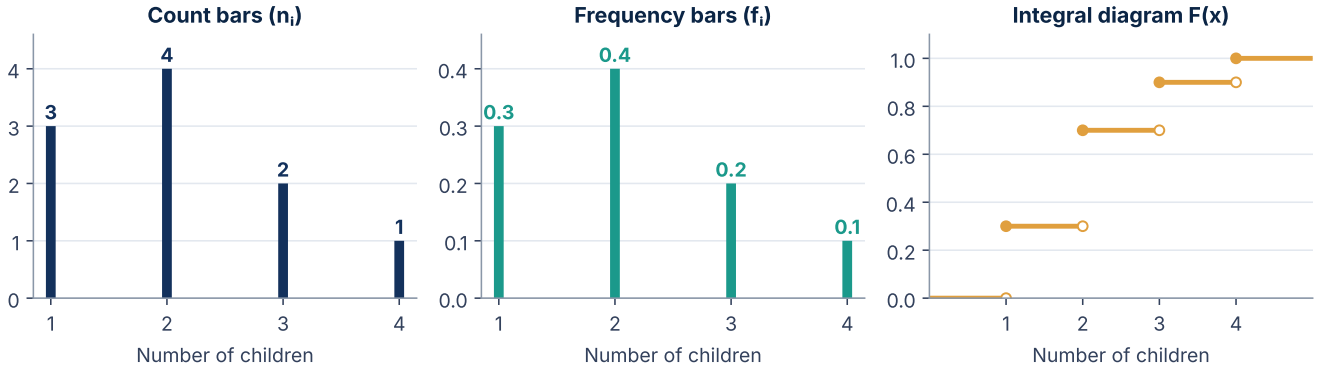
Slide 28 | Solved with the course method

COURSE REMINDER

A **discrete** variable has two graphs:

- **Bar chart** (differential diagram): one vertical bar for each value x_i, with height n_i (counts) or f_i (frequencies).
- **Cumulative graph** (integral diagram): the graph of $F(x)$. It is a **step function** (staircase) that jumps by f_i at each value x_i.

SOLUTION



- **Bar charts (differential diagram):** one vertical bar per value x_i, with height n_i or f_i. The bars are thin and separated because the variable is **discrete**.
- **Integral diagram:** a **step function**. $F(x) = 0$ for $x < 1$, then 0.3 on [1,2[, 0.7 on [2,3[, 0.9 on [3,4[and 1 for $x \geq 4$. It jumps by f_i at each value.

```
R: Freq=V/sum(V); plot(Freq, type="h", lwd=3) · plot(T, type="h", lwd=3) · plot(ecdf(X))
```

RULES TO REMEMBER

- Discrete → **bar chart** (thin separated bars) + **staircase** cumulative graph.
- Each step of the staircase is closed on the left (●) and open on the right (○). The jump at x_i equals f_i.
- The staircase starts at 0 and ends at 1.

13 Heights of 30 students (continuous variable)

Slide 32 – 33

APPLICATION

Heights (cm): 152 152 152 153 153 154 154 154 154 155 153 155 160 161 160 160 163 165 161 161 161 165 165 157 157 159 163 164 164 165.
1. Group into equal-width classes of 4 cm. 2. Create the frequency table. 3. Plot the histogram, polygon and cumulative curve.

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 13

13 Heights of 30 students (continuous variable)

Slide 32 – 33 | Solved with the course method

COURSE REMINDER

A **continuous** variable can take infinitely many values, so the data are grouped into **classes** $[e_{i-1}, e_i[$.

- Class center $c_i = \dfrac{e_{i-1} + e_i}{2}$ and class width $a_i = e_i - e_{i-1}$.
- Graphs: the **histogram** (touching rectangles), the **polygon** (joins the centers of the tops of the bars) and the **cumulative curve** (shows the cumulative distribution).

SOLUTION

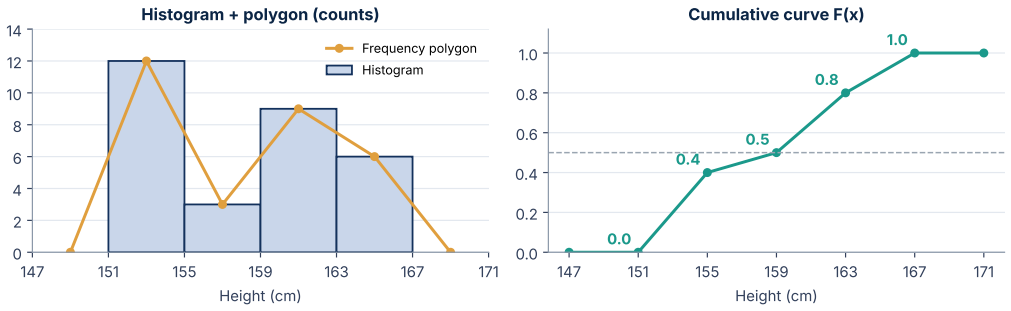
1. Classes. Min = 152 and max = 165. The professor's R code uses `breaks = c(151,155,159,163,167)`, which gives 4 classes of width $a = 4$. R's `cut()` closes the classes on the right, so they are written]151,155],]155,159],]159,163],]163,167].

2. Frequency table (count each value: 152×3, 153×3, 154×4, 155×2 → 12 in the first class, etc.):

Class	Center c_i	n_i	f_i	$n_i^c \uparrow$	$F^C \uparrow$	$F^D \downarrow$
]151, 155]	153	12	0.4	12	0.4	0.6
]155, 159]	157	3	0.1	15	0.5	0.5
]159, 163]	161	9	0.3	24	0.8	0.2
]163, 167]	165	6	0.2	30	1.0	0.0
Total		**30**	**1.0**			

Center $c_i = (e_{i-1} + e_i)/2$, e.g. $(151 + 155)/2 = 153$. All widths are equal (4 cm), so no correction is needed.

3. Graphs



The histogram uses height n_i (equal widths). The polygon joins the class centers and is closed at 149 and 169. The cumulative curve plots F at the upper bounds: (151, 0), (155, 0.4), (159, 0.5), (163, 0.8), (167, 1). It crosses 0.5 at 159, which is the median.

RULES TO REMEMBER

- Continuous → group into classes $[e_{i-1}, e_i[$. Center $c_i = \dfrac{e_{i-1} + e_i}{2}$, width $a_i = e_i - e_{i-1}$.
- Histogram = **touching** rectangles. Use corrected frequencies when the widths are unequal.
- Cumulative curve: points at the **upper bounds**, joined by segments (not a staircase).
- The polygon joins the class centers.

14 Grade obtained (qualitative ordinal)

Slide 35 – 36

APPLICATION

Variable X = grade obtained by 10 students: fair (3), good (4), very good (3). Complete the column $f_i = n_i/n$ and draw the graphs.

YOUR WORK

Solution on the next page →

14

SOLUTION · APPLICATION 14

Grade obtained (qualitative ordinal)

Slide 35 – 36 | Solved with the course method

COURSE REMINDER

A **qualitative** variable is nominal (no order, e.g. nationality) or ordinal (ordered, e.g. grades). Its table contains the modalities, the counts n_i and the relative frequencies f_i.

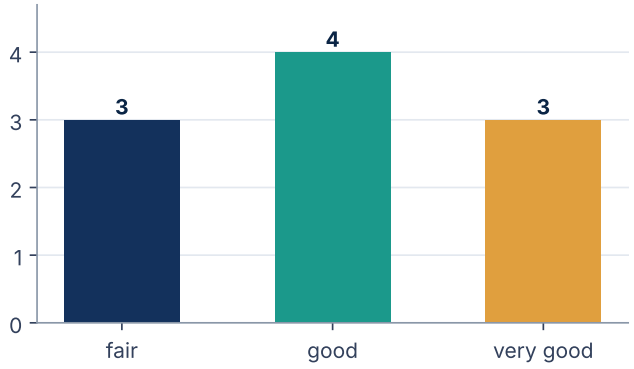
Its graphs are the **bar chart** (organ pipes) and the **pie chart** (sector diagram). In a pie chart, each category gets the angle

$$a_i = f_i \times 360^\circ$$

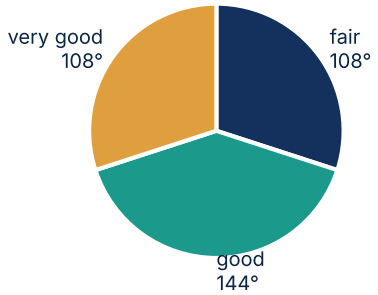
SOLUTION

X	n_i	$f_i = n_i/n$	Angle $a_i = f_i \times 360^\circ$
fair	3	0.3	108°
good	4	0.4	144°
very good	3	0.3	108°
Total	**10**	**1.0**	**360°**

Bar chart (organ pipes)



Pie chart ($a_i = f_i \times 360°$)




```
R: X=c('fair',…,'very good'); T=table(X); V=c(T); data.frame(Eff=V, freq=V/sum(V)); barplot(T);
pie(T, radius=1)
```

RULES TO REMEMBER

- Qualitative → **bar chart (organ pipes)** or **pie chart (sector diagram)**.
- Pie angle $a_i = f_i \times 360^\circ$, and the angles must add up to 360°.
- For an **ordinal** variable, keep the natural order on the axis (fair → good → very good).
- Cumulative frequencies only make sense when there is an order (ordinal or quantitative), not for nominal variables.

15 True or False: presentation of series

Slide 37

APPLICATION

Mark each statement as TRUE or FALSE.

YOUR WORK

Solution on the next page →

15

SOLUTION · APPLICATION 15

True or False: presentation of series

Slide 37 | Solved with the course method

COURSE REMINDER

The table and the graph depend on the **type of variable**:

Variable	Table	Graphs
Qualitative	modalities, n_i, f_i	bar chart (organ pipes), pie chart
Discrete	values x_i, n_i, f_i, cumulative	bar chart + cumulative staircase
Continuous	classes, c_i, n_i, f_i, cumulative	histogram, polygon, cumulative curve

SOLUTION

#	Statement	Answer	Why
1	A characteristic can only be studied either as discrete or as continuous.	FALSE	The same series can be shown in both forms. Application 16 below presents the grades in discrete *and* continuous form.
2	The frequency is the numerical expression of the characteristic.	FALSE	The numerical expression is the **value or modality** x_i. The frequency is **how many** units take it (n_i) or the proportion (f_i).
3	A statistical unit can include one or several modalities.	FALSE	Incompatibility rule: **one** modality per unit for a given variable.
4	It is preferable to conduct a sample survey rather than a census to obtain a significant study at lower cost.	TRUE	A sample is less expensive and quicker, provided it is **representative**.
5	The representation of continuous variables is a bar chart.	FALSE	Continuous variables use a **histogram**, a polygon and a cumulative curve. Bar charts are for discrete or qualitative variables.
6	In a pie chart, it is easy to represent either a discrete or a continuous variable.	FALSE	The pie chart is the graph for **qualitative** variables. Continuous classes need a histogram.

RULES TO REMEMBER

- **Graph by type:** qualitative → bars / pie · discrete → bar chart + staircase · continuous → histogram + polygon + cumulative curve.
- Value x_i ≠ count n_i ≠ frequency f_i. Do not confuse them.

16 Grades of 22 IHEC students: discrete and continuous form

Slide 38

APPLICATION

Grades: 10, 16, 16, 12, 14, 14, 16, 20, 11, 10, 10, 8, 8, 7, 7, 15, 12, 13, 18, 17, 9, 9.
1. Build the table in **discrete** form with ascending and descending cumulative frequencies. 2. Present the same series in **continuous** form with the cumulative frequencies.

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 16

16 Grades of 22 IHEC students: discrete and continuous form

Slide 38 | Solved with the course method

COURSE REMINDER

The same data can be presented in **two forms**:

- **Discrete form**: one row for each distinct value x_i, with its count. It is exact, but the table gets long when there are many values.
- **Continuous form**: the values are grouped into classes $[e_{i-1}, e_i[$. The table is shorter, but some detail is lost.

In both forms we add f_i and the ascending ($\uparrow$) and descending ($\downarrow$) cumulative frequencies.

SOLUTION

Step 0: sort the data ($N = 22$): 7 7 8 8 9 9 10 10 10 11 12 12 13 14 14 15 16 16 16 17 18 20

1. Discrete form

x_i	n_i	f_i	$n_i^c \uparrow$	$n_i^D \downarrow$	$F^C \uparrow$	$F^D \downarrow$
7	2	0.0909	2	20	0.0909	0.9091
8	2	0.0909	4	18	0.1818	0.8182
9	2	0.0909	6	16	0.2727	0.7273
10	3	0.1364	9	13	0.4091	0.5909
11	1	0.0455	10	12	0.4545	0.5455
12	2	0.0909	12	10	0.5455	0.4545
13	1	0.0455	13	9	0.5909	0.4091
14	2	0.0909	15	7	0.6818	0.3182
15	1	0.0455	16	6	0.7273	0.2727
16	3	0.1364	19	3	0.8636	0.1364
17	1	0.0455	20	2	0.9091	0.0909
18	1	0.0455	21	1	0.9545	0.0455
20	1	0.0455	22	0	1.0000	0.0000
Total	**22**	**1.0000**				

2. Continuous form. The classes are not given, so we choose **equal width 4** from 6 (min 7, max 20 covered).

Class	c_i	n_i	f_i	$n_i^c \uparrow$	$n_i^D \downarrow$	$F^C \uparrow$	$F^D \downarrow$
[6, 10[	8	6	0.2727	6	16	0.2727	0.7273
[10, 14[	12	7	0.3182	13	9	0.5909	0.4091
[14, 18[	16	7	0.3182	20	2	0.9091	0.0909
[18, 22[	20	2	0.0909	22	0	1.0000	0.0000
Total		**22**	**1.0000**				

Other classes (e.g. width 5) are fine: same method. Note that 10 goes in [10,14[because the right bound is open.

RULES TO REMEMBER

- Always **sort** the data first. Counting is then fast and safe.
- Discrete form: one row per distinct value. Continuous form: one row per class $[e_{i-1}, e_i[$.
- $n_i^D = N - n_i^c$ and $F^D = 1 - F^C$. The last n_i^c must equal N.
- A value equal to a bound goes in the class where that bound is **closed** ([ ).

PART 3 · SLIDES 39 – 58

Characteristics of a Single-Character Distribution

Central tendency (mode, median, mean), dispersion (range, MAD, variance, σ, CV) and shape (skewness, kurtosis).

17 Mode, discrete case

Slide 41

APPLICATION

x_i (children)	n_i	f_i
0	2	0.2
1	2	0.2
2	4	**0.4**
3	2	0.2

Find the mode.

YOUR WORK

Solution on the next page →

17

SOLUTION · APPLICATION 17

Mode, discrete case

Slide 41 | Solution from the slides

COURSE REMINDER

Measures of central tendency (also called position parameters) describe the center of the data. They are the mode, the median, the quantiles and the means.

The **mode** (Mo) is the value (or class) with the **highest count or frequency**: the most frequent observation. On a bar chart, it is the tallest bar.

SOLUTION

The highest relative frequency is 0.4, for $x = 2$. So **Mo = 2 children per family**.

Graphically, the mode is the **tallest bar** of the bar chart.

RULES TO REMEMBER

- Mode = the value with the **highest count or frequency**. Read it directly from the table.
- The mode is a **value** x_i (here 2), not the frequency (0.4).
- A series can have two modes (bimodal) if two values tie.

18 Mode, continuous case (interpolation)

Slide 42 – 43

APPLICATION

Class	n_i	f_i
[500–600[	3	0.1
[600–700[	6	0.2
[700–800[	18	**0.6**
[800–900[	3	0.1

YOUR WORK

Solution on the next page →

18

SOLUTION · APPLICATION 18

Mode, continuous case (interpolation)

Slide 42 – 43 | Slides + full steps added

COURSE REMINDER

For a continuous variable, we first find the **modal class**:

- with **equal** class widths: the class with the highest frequency;
- with **unequal** widths: the class with the highest **density** $d_i = \dfrac{n_i}{a_i}$ (count ÷ width).

Then the mode is located inside it by **interpolation**: $Mo = x_{inf} + a\,\dfrac{d_1}{d_1 + d_2}$, where x_{inf} is the lower bound, a the width, d_1 = f(modal) − f(previous) and d_2 = f(modal) − f(next).

SOLUTION

Step 1: modal class. All widths are equal (100), so take the highest frequency → **[700, 800[**.

Step 2: the elements. $x_{inf} = 700$, $a = 100$, $d_1 = 0.6 - 0.2 = 0.4$ (gap with the previous class), $d_2 = 0.6 - 0.1 = 0.5$ (gap with the next class).

Step 3: formula.

$$Mo = x_{inf} + a\,\frac{d_1}{d_1 + d_2} = 700 + 100 \times \frac{0.4}{0.4 + 0.5} = 700 + 44.4 = \mathbf{744.4}$$

The mode is pulled towards [600,700[because $d_1 < d_2$: the previous class is more populated than the next one.

RULES TO REMEMBER

- $Mo = x_{inf} + a\,\dfrac{d_1}{d_1 + d_2}$, with d_1 = modal − **previous** and d_2 = modal − **next**.
- **Equal** widths → modal class = highest frequency. **Unequal** widths → highest **density** $d_i = n_i / a_i$.
- You can use n_i or f_i for d_1 and d_2; the ratio is the same.
- Check: the mode must fall **inside** the modal class.

19 Median, discrete data (odd and even n)

Slide 45

APPLICATION

Series 1: 14, 15, 9, 18, 10, 15, 19, 12, 13, 14. Series 2: add one more value (10). Find the median of each.

YOUR WORK

Solution on the next page →

19

SOLUTION · APPLICATION 19

Median, discrete data (odd and even n)

Slide 45 | Solved with the course method

COURSE REMINDER

The **median** (Me) divides the **ordered** data into two equal parts: 50 % of the values are below it and 50 % are above. So $F(Me) = 0.5$ and $N(Me) = n/2$.

- Odd number of observations ($n = 2m + 1$): Me = the value in position $m + 1$.
- Even number ($n = 2m$): Me = the average of the values in positions m and $m + 1$.

SOLUTION

Series 1. Sorted: 9, 10, 12, 13, **14, 14**, 15, 15, 18, 19. Here $n = 10 = 2m$, so $m = 5$ (even case):

$$Me = \frac{x_{(5)} + x_{(6)}}{2} = \frac{14 + 14}{2} = \mathbf{14}$$

Series 2. Sorted: 9, 10, 10, 12, 13, **14**, 14, 15, 15, 18, 19. Here $n = 11 = 2m + 1$, so $m = 5$ (odd case):

$$Me = x_{(m+1)} = x_{(6)} = \mathbf{14}$$

Interpretation: 50 % of the values are ≤ 14 and 50 % are ≥ 14.

RULES TO REMEMBER

- **Always sort first**; the median uses positions in the ordered list.
- n odd ($2m + 1$): Me = the value in position $m + 1$ (the middle one).
- n even ($2m$): Me = the average of the values in positions m and $m + 1$.
- Position shortcut: the middle is at position $\frac{n+1}{2}$.

20 Median, continuous data (interpolation)

Slide 46

APPLICATION

Class	n_i
[600–800[	5
[800–1000[	13
[1000–1200[	19
[1200–1400[	3
Total	**40**

Compute the median.

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 20

20 Median, continuous data (interpolation)

Slide 46 | Solved with the course method

COURSE REMINDER

For grouped (continuous) data we cannot read the median directly, so we:

1. find the **median class**: the first class where the cumulative frequency reaches 0.5;
2. apply **linear interpolation** inside it:

$$Me = x_i + a_i \frac{0.5 - F(x_i)}{F(x_{i+1}) - F(x_i)}$$

where x_i is the lower bound of the median class and a_i its width.

SOLUTION

Class	n_i	f_i	F^C (at upper bound)
[600, 800[	5	0.125	0.125
[800, 1000[	13	0.325	0.450
[1000, 1200[	19	0.475	**0.925**
[1200, 1400[	3	0.075	1.000
Total	**40**	**1.000**	

Median class: the first class where F reaches 0.5 → **[1000, 1200[** (F goes from 0.45 to 0.925).

$$Me = x_i + a_i \frac{0.5 - F(x_i)}{F(x_{i+1}) - F(x_i)} = 1000 + 200 \times \frac{0.5 - 0.45}{0.925 - 0.45} = 1000 + 200 \times \frac{0.05}{0.475} \approx \mathbf{1021.05}$$

Interpretation: half of the employees earn less than about 1021, and half earn more.

With counts, the same calculation is $N/2 = 20$: $Me = 1000 + 200 \times \frac{20 - 18}{19} = 1021.05$.

RULES TO REMEMBER

- Median class = first class where $F^C \geq 0.5$ (or $n_i^c \geq N/2$).
- x_i = **lower bound** of the median class, $F(x_i)$ = cumulative frequency **before** it, and $F(x_{i+1}) - F(x_i) = f_i$ of the median class.
- Check: the median must fall inside the median class.
- Graphically, the median is where the cumulative curve crosses 0.5.

21 Mean absolute deviation (salary distribution)

Slide 49 – 50

APPLICATION

Class	n_i
[600–800[	5
[800–1000[	13
[1000–1200[	19
[1200–1400[	3
Total	**40**

Compute the mean absolute deviation e_m (grouped data → use the class midpoints).

YOUR WORK

Solution on the next page →

21

SOLUTION · APPLICATION 21

Mean absolute deviation (salary distribution)

Slide 49 – 50 | Solved with the course method

COURSE REMINDER

Central tendency (mean, median, mode) does not tell us **how spread out** the data are. **Measures of dispersion** show how close or far the values are from the center. They let us compare datasets with the same mean but different variability.

The **mean absolute deviation** is the average distance to the mean:

$$e_m = \frac{1}{n}\sum n_i|x_i - \bar{X}| = \sum f_i|x_i - \bar{X}|, \qquad \bar{X} = \frac{1}{n}\sum n_i x_i$$

For grouped data, x_i is the class midpoint. A variant measures the distance to the median: e_{Me}.

SOLUTION

Step 1: centers and mean.

$$\bar{X} = \frac{1}{n}\sum n_i c_i = \frac{5(700) + 13(900) + 19(1100) + 3(1300)}{40} = \frac{40\,000}{40} = \mathbf{1000}$$

Class	c_i	n_i	$\lvert c_i - \bar{X}\rvert$	$n_i\lvert c_i - \bar{X}\rvert$
[600, 800[	700	5	300	1500
[800, 1000[	900	13	100	1300
[1000, 1200[	1100	19	100	1900
[1200, 1400[	1300	3	300	900
Total		**40**		**5600**

Step 2:

$$e_m = \frac{1}{n}\sum n_i|c_i - \bar{X}| = \frac{5600}{40} = \mathbf{140}$$

On average, salaries are 140 away from the mean.

Alternative (slide 50): MAD around the median ($Me \approx 1021.05$ from Application 20):

$$e_{Me} = \frac{5(321.05) + 13(121.05) + 19(78.95) + 3(278.95)}{40} = \frac{5515.8}{40} \approx \mathbf{137.9}$$

RULES TO REMEMBER

- $e_m = \frac{1}{n}\sum n_i|x_i - \bar{X}| = \sum f_i|x_i - \bar{X}|$. Grouped data → replace x_i by the center c_i.
- Mean: $\bar{X} = \frac{1}{n}\sum n_i x_i = \sum f_i x_i$.
- Advantage: it uses all values. Drawback: absolute values are hard to handle in algebra, which is why we use the variance.

22 Variance and standard deviation (salary distribution)

Slide 51 – 52

APPLICATION

Compute the variance and standard deviation of the same salary distribution (classes [600–800[… [1200–1400[, n_i = 5, 13, 19, 3).

YOUR WORK

Solution on the next page →

22

SOLUTION · APPLICATION 22

Variance and standard deviation (salary distribution)

Slide 51 – 52 | Solved with the course method

COURSE REMINDER

The absolute values in the MAD are hard to handle in algebra. **Variance** solves this by **squaring** the deviations, which removes the sign. It is the average squared deviation from the mean (also called the *mean square deviation*):

$$V(X) = \frac{1}{n}\sum n_i(x_i - \bar{X})^2 = \left(\frac{1}{n}\sum n_i x_i^2\right) - \bar{X}^2$$

The **standard deviation** $\sigma = \sqrt{V(X)}$ expresses the dispersion in the **same unit** as the variable. The greater σ is, the more the values deviate from the mean.

SOLUTION

Use the **computational formula** $V(X) = \left(\frac{1}{n}\sum n_i c_i^2\right) - \bar{X}^2$, with $\bar{X} = 1000$:

c_i	n_i	c_i^2	$n_i c_i^2$
700	5	490 000	2 450 000
900	13	810 000	10 530 000
1100	19	1 210 000	22 990 000
1300	3	1 690 000	5 070 000
Total	**40**		**41 040 000**

$$V(X) = \frac{41\,040\,000}{40} - 1000^2 = 1\,026\,000 - 1\,000\,000 = \mathbf{26\,000}$$

Check with the definition: $\frac{1}{40}\left[5(300^2) + 13(100^2) + 19(100^2) + 3(300^2)\right] = \frac{1\,040\,000}{40} = 26\,000$ ✓

$$\sigma = \sqrt{V(X)} = \sqrt{26\,000} \approx \mathbf{161.25}$$

σ is in the **same unit** as the salary. Bonus: $CV = \sigma/\bar{X} = 161.25/1000 \approx 16.1\,\%$.

RULES TO REMEMBER

- $V(X) = \frac{1}{n}\sum n_i(x_i - \bar{X})^2 = \overline{X^2} - \bar{X}^2$ ("mean of the squares minus the square of the mean").
- $\sigma = \sqrt{V}$. A variance is **never negative**; a negative result means a calculation error.
- The variance is in unit², and σ is in the unit of the variable.
- Grouped data → use the centers c_i.

23 Coefficient of variation: comparing two stores

Slide 53 – 54

APPLICATION

Store 1 profit	n_i	Store 2 profit	n_i
1000	10	11000	8
2000	25	12000	24
3000	37	13000	38
4000	21	14000	20
5000	7	15000	10
Total	**100**		**100**

Compare the variability of the weekly net profits of the two stores (100 weeks).

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 23

23 Coefficient of variation: comparing two stores

Slide 53 – 54 | Slides + full steps added

COURSE REMINDER

Problem: the variance and σ depend on the measurement unit and on the size of the values, so they cannot compare series with very different means.

Solution: the **coefficient of variation**, a unit-free measure of **relative** dispersion:

$$CV(X) = \frac{\sigma(X)}{\bar{X}}$$

SOLUTION

Store 1. $\sum n_i x_i = 290\,000 \rightarrow \bar{X}_1 = 2900.$ $\sum n_i x_i^2 = 954\,000\,000 \rightarrow \overline{X^2} = 9\,540\,000.$
$V_1 = 9\,540\,000 - 2900^2 = 9\,540\,000 - 8\,410\,000 = 1\,130\,000 \rightarrow \sigma_1 \approx 1063.$

Store 2. $\sum n_i x_i = 1\,300\,000 \rightarrow \bar{X}_2 = 13\,000.$ $\sum n_i x_i^2 = 17\,016\,000\,000 \rightarrow \overline{X^2} = 170\,160\,000.$
$V_2 = 170\,160\,000 - 13\,000^2 = 1\,160\,000 \rightarrow \sigma_2 \approx 1077.$

Statistic	Store 1	Store 2
Mean $\bar{X}$	2 900	13 000
σ	1 063	1 077
$CV = \sigma/\bar{X}$	**36.65 %**	**8.28 %**

Interpretation (from the slides): the two σ are almost the same, but Store 1's profits are much **more variable relative to its mean**. The CV captures this relative dispersion.

RULES TO REMEMBER

- $CV = \frac{\sigma}{\bar{X}}$ has **no unit** (often written as %). Use it to compare series with different means or units.
- Higher CV → more relative dispersion → less homogeneous.
- Similar σ does **not** mean similar variability when the means are very different.

24 Bonus: shape of the salary distribution

Slide 55 – 58

APPLICATION

This application is **built from course data**: slides 55–58 have no exercise, so we apply them to the salary distribution of slides 46–51. Determine its **skewness** (compare $\bar{X}$, Me, Mo) and its **kurtosis** (Pearson's A_p).

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 24

24 Bonus: shape of the salary distribution

Slide 55 – 58 | Solved with the course method

COURSE REMINDER

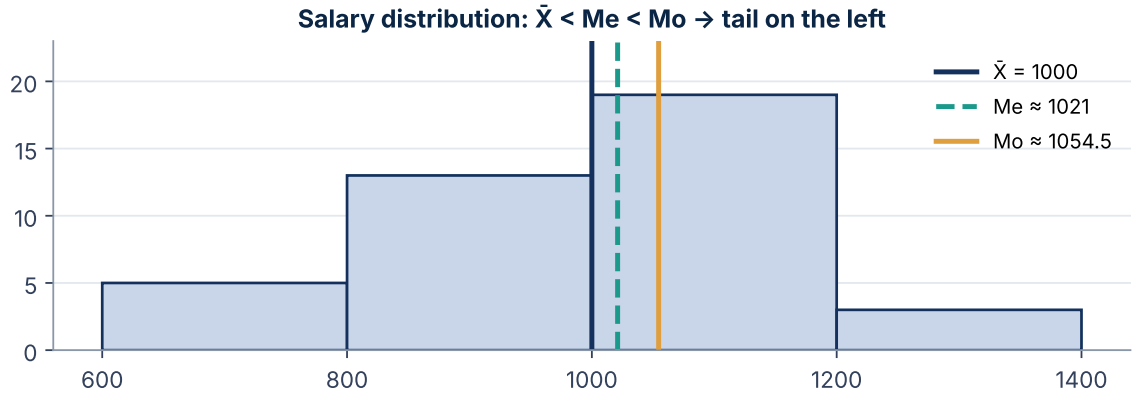
Shape characteristics describe the form of a distribution without drawing it:

- **Asymmetry (skewness)**: compare the mean, median and mode. $Mo < Me < \bar{X}$ means right-skewed, $Mo = Me = \bar{X}$ means symmetric, and $Mo > Me > \bar{X}$ means left-skewed.
- **Flatness (kurtosis)**: how peaked or flat the curve is compared with the normal curve. It is measured by Pearson's coefficient $A_p = \frac{\mu_4}{\mu_2^2} = \frac{\mu_4}{\sigma^4}$. $A_p = 3$ is mesokurtic, $A_p > 3$ leptokurtic (more peaked) and $A_p < 3$ platykurtic (flatter).

SOLUTION

1. The three central values. $\bar{X} = 1000$ (App. 21) and $Me \approx 1021.05$ (App. 20).
Mode: equal widths (200), so the modal class is [1000, 1200[($f = 0.475$). $d_1 = 0.475 - 0.325 = 0.15$ and $d_2 = 0.475 - 0.075 = 0.40$:

$$Mo = 1000 + 200 \times \frac{0.15}{0.15 + 0.40} \approx \mathbf{1054.5}$$



Skewness: $Mo > Me > \bar{X}$ (1054.5 > 1021 > 1000) → **left-skewed** distribution (tail on the left). The few low salaries pull the mean down.

2. Kurtosis. Fourth central moment, using the centers:

$$\mu_4 = \frac{1}{n}\sum n_i(c_i - \bar{X})^4 = \frac{5(300^4) + 13(100^4) + 19(100^4) + 3(300^4)}{40} = \frac{68 \times 10^9}{40} = 1.7 \times 10^9$$

$$A_p = \frac{\mu_4}{\sigma^4} = \frac{1.7 \times 10^9}{26\,000^2} = \frac{1.7 \times 10^9}{6.76 \times 10^8} \approx \mathbf{2.51} < 3$$

→ **Platykurtic**: the distribution is flatter than the normal curve.

RULES TO REMEMBER

- $Mo < Me < \bar{X}$ → right-skewed (tail right). $Mo = Me = \bar{X}$ → symmetric. $Mo > Me > \bar{X}$ → left-skewed (tail left).
- Memory trick: the **mean runs towards the tail**.
- $A_p = \mu_4/\sigma^4$ (with $\mu_2 = \sigma^2$): = 3 mesokurtic (normal) · > 3 leptokurtic (more peaked) · < 3 platykurtic (flatter).

PART 4 · SLIDES 59 – 75

Probability Distributions: Discrete Laws

Bernoulli, Binomial and Poisson: recognise the law, give its parameters, compute probabilities, expectation and variance.

25 Bernoulli: vehicle inspection center

Slide 62 – 63

APPLICATION

At a vehicle inspection center, X = 1 if a car passes and X = 0 if it fails. The rejection rate is 0.1.
1. What probability distribution does X follow? Give its parameters. 2. Compute E(X) and V(X).

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 25

25 Bernoulli: vehicle inspection center

Slide 62 – 63 | Solution from the slides

COURSE REMINDER

A **random variable** associates a number with each outcome of a random experiment. It is **discrete** if it takes countable values (0, 1, 2, ...).

The **Bernoulli distribution** $\mathcal{B}(p)$ models **one** trial with **two opposite outcomes**: success (X = 1) with probability p and failure (X = 0) with probability $q = 1 - p$:

$$p(X = x) = p^x(1-p)^{1-x},\ x \in \{0,1\}, \qquad E(X) = p, \quad V(X) = p(1-p)$$

SOLUTION

1. The experiment has two opposite results: **success** (the car is accepted) with probability 0.9, and **failure** (the car is rejected) with probability 0.1. So X follows a **Bernoulli distribution with parameter p = 0.9**:

$$X \rightsquigarrow \mathcal{B}(0.9), \qquad p(X = x) = \begin{cases} (0.9)^x(0.1)^{1-x} & \text{if } x = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

2. $E(X) = p = \mathbf{0.9}$ $\quad V(X) = pq = (0.9)(0.1) = \mathbf{0.09}$ (so $\sigma = 0.3$).

RULES TO REMEMBER

- Bernoulli = **one** trial with **two** outcomes (1 = success, 0 = failure).
- p = probability of the event **coded 1**. Be careful: here “rejection 0.1” gives p = 0.9 because 1 = pass.
- $E(X) = p$ and $V(X) = p(1-p) = pq$.
- The sum of two independent Bernoulli variables is **not** Bernoulli (it can be 2).

26 Binomial: identify the distributions

Slide 66

APPLICATION

Determine the binomial distribution from each representation:

1. $p(X = x) = C_{20}^{x}(0.2)^{x}(0.8)^{20-x}$ for $x = 0, \ldots, 20$
2. $p(X = x) = C_{10}^{x}(0.3)^{x}(0.7)^{10-x}$ for $x = 0, \ldots, 10$
3. $p(X = x) = C_{10}^{x}(0.4)^{x}(0.5)^{10-x}$ for $x = 0, \ldots, 10$
4. $p(X = x) = C_{20}^{x}(0.4)^{x}(0.6)^{10-x}$ for $x = 0, \ldots, n$

YOUR WORK

Solution on the next page →

SOLUTION · APPLICATION 26

26 Binomial: identify the distributions

Slide 66 | Solved with the course method

COURSE REMINDER

The **binomial distribution** $\mathcal{B}(n,p)$: repeat n **independent** Bernoulli trials, each with the same success probability p, and let X be the **number of successes**:

$$p(X=x)=C_n^x\,p^x(1-p)^{n-x},\quad x=0,1,\dots,n,\qquad C_n^x=\frac{n!}{x!(n-x)!}$$

Properties: $E(X)=np$, $V(X)=npq$, and $\mathcal{B}(1,p)=\mathcal{B}(p)$.

SOLUTION

Compare each expression with the model $C_n^x\,p^x(1-p)^{n-x}$, $x=0,\dots,n$. Check that $p+q=1$ and that the same n appears everywhere.

#	Check	Conclusion
1	n = 20, p = 0.2, q = 0.8, and 0.2 + 0.8 = 1 ✓	$X \rightsquigarrow \mathcal{B}(20;\,0.2)$: E = np = 4, V = npq = 3.2
2	n = 10, p = 0.3, q = 0.7, and 0.3 + 0.7 = 1 ✓	$X \rightsquigarrow \mathcal{B}(10;\,0.3)$: E = 3, V = 2.1
3	0.4 + 0.5 = 0.9 ≠ 1 ×	**Not a binomial**: the probabilities do not sum to 1
4	C_{20}^x but exponent $10-x$, and x goes to an unspecified n ×	**Not a binomial**: n is not the same in C_n^x and in $(1-p)^{n-x}$

RULES TO REMEMBER

- Binomial $\mathcal{B}(n,p)$: $p(X=x)=C_n^x p^x(1-p)^{n-x}$ for $x=0,1,\dots,n$, with $C_n^x=\dfrac{n!}{x!(n-x)!}$.
- Three checks: **p + q = 1**; the **same n** in the combination and in the exponent; x goes from 0 to n.
- $E(X)=np$, $V(X)=npq$. Also $\mathcal{B}(1,p)=\mathcal{B}(p)$.

27 Binomial: ORANGE customers

Slide 67

APPLICATION

6 % of Tunisians are customers of the operator “ORANGE”. 50 Tunisians are surveyed; X = number of ORANGE customers among them.
1. Determine, with justification, the distribution of X. 2. Probability that the 6 % proportion is observed? 3. Probability that none is an ORANGE customer? 4. Probability of at least four ORANGE customers? 5. Probability of more than 10 ORANGE customers?

YOUR WORK

Solution on the next page →

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SOLUTION · APPLICATION 27

Binomial: ORANGE customers

Slide 67 | Solved with the course method

COURSE REMINDER

To recognise a binomial situation, look for: a **fixed number** n of trials, each with **two outcomes** (success or failure), the **same probability** p every time, and **independent** trials. X counts the successes, and $X \rightsquigarrow \mathcal{B}(n,p)$.

Useful tool: the complement $P(X \geq k) = 1 - P(X \leq k-1)$. **Approximation (slide 73):** if $n > 30$, $p < 0.1$ and $np < 10$, then $\mathcal{B}(n,p) \approx \mathcal{P}(\lambda = np)$.

SOLUTION

1. Each person is a Bernoulli trial: success = "is an ORANGE customer" with $p = 0.06$. The 50 trials are **independent** with the **same p**, and X counts the successes. So

$$X \rightsquigarrow \mathcal{B}(50;\, 0.06), \quad p(X = x) = C_{50}^{x}(0.06)^{x}(0.94)^{50-x},\ x = 0, \ldots, 50$$

($E(X) = np = 3$ and $V(X) = npq = 2.82$.)

2. 6 % of 50 = 3 customers: $P(X = 3) = C_{50}^{3}(0.06)^{3}(0.94)^{47} = 19600 \times 0.000216 \times 0.05458 \approx \mathbf{0.2311}$

3. $P(X = 0) = (0.94)^{50} \approx \mathbf{0.0453}$

4. Use the complement: $P(X \geq 4) = 1 - P(X \leq 3) = 1 - [P(0) + P(1) + P(2) + P(3)]$
$= 1 - (0.0453 + 0.1447 + 0.2262 + 0.2311) = 1 - 0.6473 \approx \mathbf{0.3527}$

5. $P(X > 10) = 1 - P(X \leq 10) \approx \mathbf{0.00015}$ (practically 0).

Check with slide 73: n = 50 > 30, p = 0.06 < 0.1 and np = 3 < 10, so $X \approx \mathcal{P}(3)$. This gives P(X=3) ≈ 0.2240, P(X=0) ≈ 0.0498, P(X≥4) ≈ 0.3528 and P(X>10) ≈ 0.0003, very close to the exact values. Use it when you only have a Poisson table.

RULES TO REMEMBER

- Justify a binomial with 4 points: **n identical trials**, **2 outcomes**, **same p**, **independent**, and X = number of successes.
- "At least k" → $1 - P(X \leq k-1)$. "More than k" → $1 - P(X \leq k)$. "None" → $P(X = 0) = q^{n}$.
- Approximation $\mathcal{B}(n,p) \approx \mathcal{P}(np)$ when n > 30, p < 0.1 and np < 10.

28 Poisson: two toll booths

Slide 74 – 75

APPLICATION

On a highway, two toll booths A and B operate. X_A and X_B are the numbers of cars served in 5 minutes, with $X_A \rightsquigarrow \mathcal{P}(3)$ and $X_B \rightsquigarrow \mathcal{P}(2)$, and the services are independent. Compute $P(X > 3)$ where $X = X_A + X_B$.

YOUR WORK

Solution on the next page →

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SOLUTION · APPLICATION 28

Poisson: two toll booths

Slide 74 – 75 | Slides + full steps added

COURSE REMINDER

The **Poisson distribution** $\mathcal{P}(\lambda)$ models the number of occurrences of a **rare and independent** event over a fixed interval of time, area or volume (phone calls, defects, customers, accidents...):

$$p(X=x)=\frac{\lambda^x e^{-\lambda}}{x!}, \quad x=0,1,2,\ldots \qquad E(X)=V(X)=\lambda$$

Stability property: if $X \rightsquigarrow \mathcal{P}(\lambda)$ and $Y \rightsquigarrow \mathcal{P}(\beta)$ are independent, then $X+Y \rightsquigarrow \mathcal{P}(\lambda+\beta)$.

SOLUTION

1. Sum of two independent Poisson variables (stability property):

$$X = X_A + X_B \rightsquigarrow \mathcal{P}(3+2) = \mathcal{P}(5)$$

2. Complement:

$$P(X>3) = 1 - P(X \leq 3) = 1 - \left[\frac{5^0 e^{-5}}{0!} + \frac{5^1 e^{-5}}{1!} + \frac{5^2 e^{-5}}{2!} + \frac{5^3 e^{-5}}{3!}\right]$$

$$= 1 - (0.0067379 + 0.0336897 + 0.0842243 + 0.1403739) = 1 - 0.265 = \mathbf{0.735}$$

Interpretation: in 5 minutes, there is a 73.5 % chance that more than 3 cars are served in total.

RULES TO REMEMBER

- Poisson $\mathcal{P}(\lambda)$: $p(X=x)=\frac{\lambda^x e^{-\lambda}}{x!}$ for $x=0,1,2,\ldots$ (no upper limit).
- $E(X)=V(X)=\lambda$ (mean = variance).
- Independent sums: $\mathcal{P}(\lambda)+\mathcal{P}(\beta)=\mathcal{P}(\lambda+\beta)$.
- Use Poisson for **rare, independent events counted over an interval** of time or space. Use Binomial for a **fixed n**.
- Shortcut: $e^{-5} \approx 0.0067$. Then each term is the previous one $\times\, \lambda/x$.